Aysin Tumay

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Research interests: Safe offline reinforcement learning for medical device automation, with a focus on uncertainty mitigation, and evaluation by digital twins.

Education

University of California, San Diego **Sep. 2024 – Jun. 2029**

Ph.D. in Computer Science and Engineering GPA: 4.00/4.00; *La Jolla, USA*

- Advisor: **Prof. Rose Yu**
- Selected coursework: Convex Optimization (CSE203B), Machine Learning Theory (CSE251C), Deep Learning (CSE251B).

Bilkent University **Sep. 2019 – Jun. 2024**

B.Sc. in Electrical and Electronics Engineering GPA: 3.80/4.00, *Rank:* 10/158 *Ankara, Turkey*

- Selected coursework: Statistical Foundations of Natural Language Processing (EE486), Linear Algebra in Data Analysis and Machine Learning (EE361).

Experience

DataBoss Security and Analytics **Dec. 2022 - May 2024**

Machine Learning Researcher *Ankara, Turkey*

∗ Developed state-of-the-art sequential models for high-dimensional time series using Gradient Boosting.

DataBoss Security and Analytics **Aug. - Sep. 2022**

Machine Learning Intern *Ankara, Turkey*

∗ Applied Machine Learning techniques on time series data using Gradient Boosting and Neural Network models with Python.

ASELSAN, Radar and Warfare Systems **Jun. - Jul. 2022**

Algorithm Design Intern *Ankara, Turkey*

∗ Implemented radar geolocation algorithms in MATLAB, improving detection accuracy.

UMRAM, Bilkent University **Aug. 2021**

Undergraduate Research Assistant *Ankara, Turkey*

∗ Practiced Deep Learning methods for detecting brain illnesses in MRI scans.

Projects

Trajectory Prediction for Autonomous Driving **Apr. - Jun. 2025**

∗ Leveraged Transformers with a mixture of experts for trajectory prediction in the spatio-temporal domain.

Image Colorization with Convex Optimization **Jan. - Mar. 2025**

∗ Designed a Convolutional Neural Network model and formulated the primal and dual solutions for art image colorization.

Recipe Recommendation with Machine Learning **Sep. - Dec. 2024**

∗ Utilized XGBoost, SVD, BERT and similarity-based models.

Spatiotemporal Traffic Accident Prediction in Turkey **Jun. 2023- May 2024**

∗ Designed machine learning models to predict the probability of traffic accidents with Neural Networks and Boosting.

∗ Conducted research on tackling data sparsity and spatial heterogeneity.

Wind Energy Production Prediction **Feb. - May 2023**

∗ Predicted hourly total electrical energy consumption in Spain with Linear Regression, Decision Tree, and AdaBoost.

A Basic Level Category Analysis with Commonsense Question Answering **Feb. - May 2023**

∗ Measured the common sense question answering performance of one of GPT-3.5-turbo on Family Feud language game.

Song Recommendation System for Spotify Playlists | *Python, TensorFlow* **Sep. - Dec. 2022**

∗ Applied unsupervised clustering algorithms such as k-Means, DBScan, and Autoencoder to give song recommendations to a playlist.

Honors & Awards

University of California, San Diego, CSE Graduate Student Researcher Fellowship **Mar. 2024**
* Full tuition waiver, and stipend for entire duration of Ph.D. studies.

University of California, San Diego, ECE Fellowship **Mar. 2024**
* Recipient of first year's financial support rewarded to exceptional PhD applicants.

University of Illinois at Urbana-Champaign, the Dilip and Sandhya Sarwate Graduate Fellowship **Mar. 2024**
* Awarded to outstanding incoming graduate students in the area of communications.

University of Illinois at Urbana-Champaign, the Promise of Excellence Fellowship **Mar. 2024**
* Given to incoming PhD students who demonstrate the potential to achieve great things in the ECE field.

3^{rd} **Place at Ipsos Datathon** | *Jupyter* **May 2023**
* Solved a case study about predicting a company's market share by trend analysis using ARIMA and Linear Regression.

5^{th} **Place at Invent Analytics Data Analysis Challange** | *Jupyter* **Sep. 2022**
* Trained and tested a Machine Learning model to forecast the sales amount of a clothing brand.

Technical Skills

Languages: Python, VHDL, MATLAB, Assembly 8051,
Tools: VSCode, Pycharm, Jupyter Notebook, Linux, GitHub, LaTex, MS applications,
AI/ML Frameworks: Gymnasium, Pytorch, Tensorflow, Scikit-learn,
Electronics: LTSpice, DipTrace, Proteus, MCU IDE.

Publications

A. Tumay, S. Sun, S. Fereidooni, A. Dumas, E. Jortberg, and R. Yu, "Guardian-regularized Safe Offline Reinforcement Learning for Smart Weaning of Mechanical Circulatory Devices," Machine Learning for Health Symposium, 2025. DOI: 10.48550/ARXIV.2511.06111.

S. H. Sun, **A. Tumay**, S. Fereidooni, A. Dumas, E. Jortberg, and R. Yu, "Probabilistic Digital Twin for Data-Driven Smart Weaning of Mechanical Circulatory Devices," ***NeurIPS 2025 Workshop on Learning from Time Series for Health***, Accepted, 2025.

A. Tumay, M. E. Aydin, A. T. Koc, S. S. Kozat. "Hierarchical Ensemble-based Feature Selection for Time Series Forecasting." 2023. DOI: 10.48550/ARXIV.2310.17544.

Professional Service

- Reviewer for ML4H symposium (2025).
- Reviewer for NeurIPS TS4H workshop (2025).

Extracurricular

- Organizing lunch events for Women in Data Science at UCSD.
- Active member at Young Entrepreneurs Society, and IEEE Student Branch, 2019-2020.
- Ankara Start-up Summit committee member, 2019-2020.
- Classical guitar player in high school orchestra, 2018.